

Leveraging Machine Learning for Advancements in IoT Systems

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What is Internet of Things (IoT)?

- A network of physical devices embedded with sensors, software, and connectivity.
- Facilitates automation and enhances decision-making through real-time data.
- 25 billion devices predicted to be connected by 2025.
- 1.1 trillion USD market by 2030.

Image source: Prepp Platform

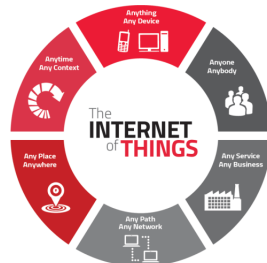


Figure: Features of IoT.

IoT Applications

- Security and Privacy
- Smart (Cities and Homes)
- Industrial Automation
- Blockchain

Security and Privacy

- **Highly security-centric** "life-and-death" applications
- **Intermediate security** uses that include: smart homes
- **Lower security** casual uses such as: games

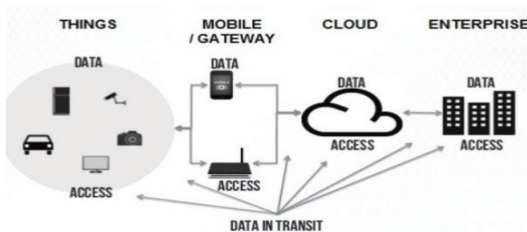


Figure: Security and privacy in IoT.

Smart (Homes and Cities)

- **Home Area Network** (Zigbee, BLE, WiFi): Smart Homes
- **Wide Area Network**: Smart Cities
- **Field Area Networks**: Smart End-to-end



Figure: Smart IoT.

Industrial Automation

Industrial IoT (IIoT)

- IoT revolutionizes industrial operations by integrating smart devices, sensors, and advanced analytics to automate processes.

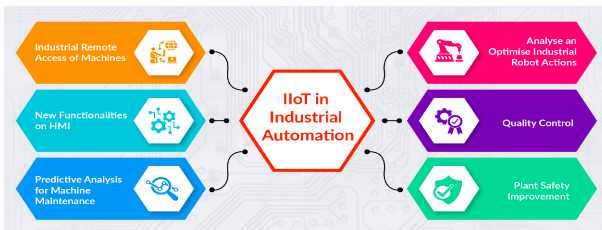


Figure: Practical applications of IIoT in industrial automation.

Blockchain

Blockchain with IoT

- Blockchain is a distributed ledger system that eliminates the need for any centralized authority and offers security and privacy.
- IoT devices use blockchain technology as the backend networking system for storing their information.



Figure: Blockchain with IoT startup.

IoT Architecture

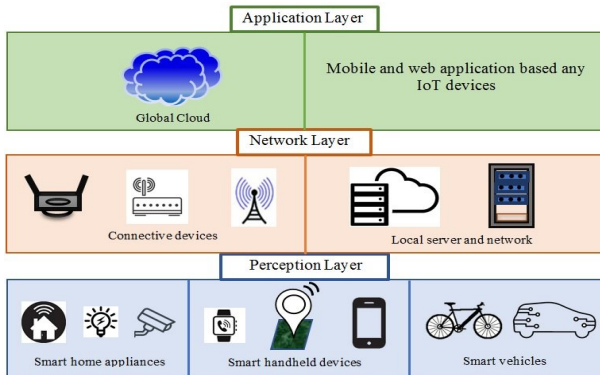


Figure: IoT architecture.

IoT Devices



Figure: Wearable and connecting devices.

Machine Learning

Tom M. Mitchell

A computer program is said to learn from **Experience (E)** with respect to some **Task (T)** and some **Performance measure (P)**, if its performance on T, as measured by P, improves with experience E.

Learning means **Improving** with **Experience** at some **Task**.

Example: Smart Homes

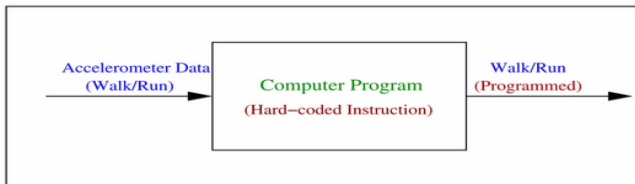
- **T**: Estimate the desired temperature
- **E**: Learning from temperature dataset
- **P**: Accuracy of the desired temperature

Why Machine Learning?

- IoT generates massive dataset requiring advanced analytics.
- Machine learning enables predictive capabilities and autonomous decision-making.
- **Advantages:**
 - Improves system efficiency.
 - Enhances real-time insights.

Traditional Programming vs Machine Learning

Traditional Programming:



Machine Learning:

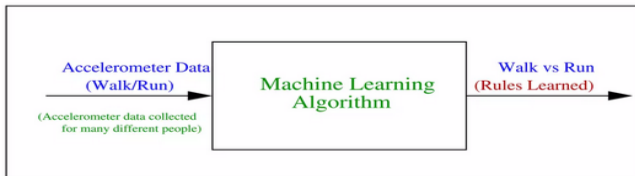
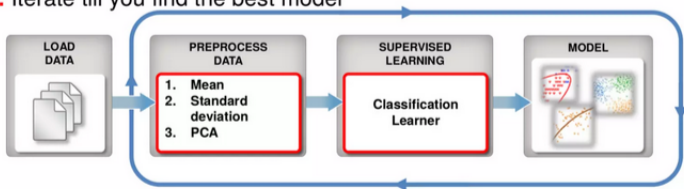


Figure: Comparison between traditional programming and machine learning.

Machine Learning (ML)

Train: Iterate till you find the best model



Predict: Integrate trained models into applications



Figure: Abstract view of machine learning.

Overview of Machine Learning Techniques

- **Supervised Learning:**

- Relies on labeled data to learn the mapping between inputs and outputs.
- Common tasks include:
 - **Regression:** Predicting continuous values.
 - **Classification:** Assigning categories to data points.

- **Unsupervised Learning:**

- Works with unlabeled data to uncover hidden structures.
- **Clustering:** Grouping similar data points.

- **Reinforcement Learning:**

- Focuses on training agents to make sequences of decisions in dynamic environments.
- The agent learns through:
 - **Rewards:** Positive feedback for correct actions.
 - **Penalties:** Negative feedback for incorrect actions.

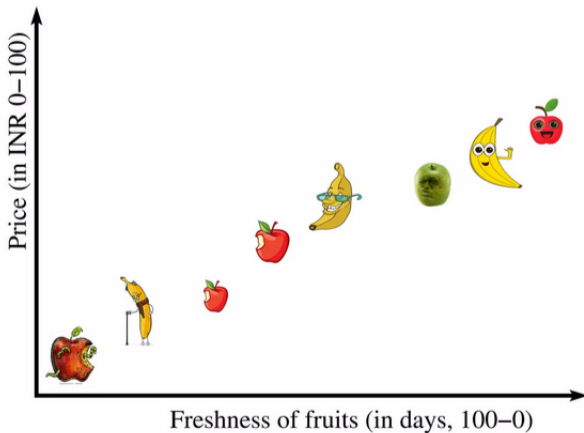
Supervised Learning: Regression

Question

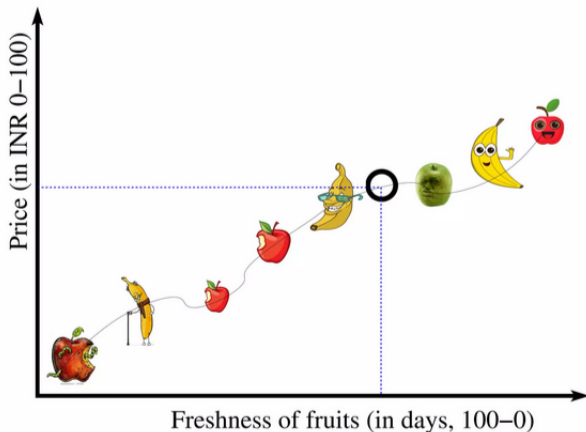
How to **estimate** the price of a fruit if freshness (in days) is given?



Supervised Learning: Regression



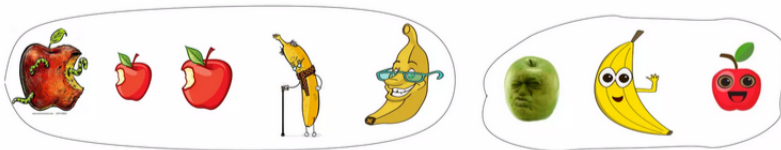
Supervised Learning: Regression



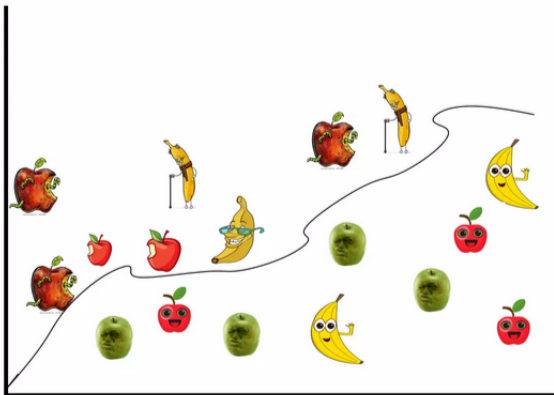
Supervised Learning: Classification

Question

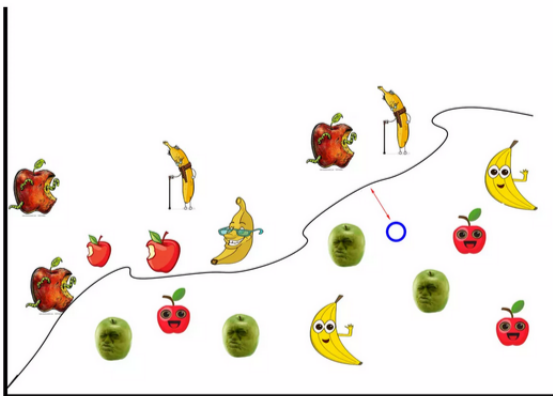
How to **identify** where a given fruit is fresh (if freshness is not more than 50 days) or not?



Supervised Learning: Classification

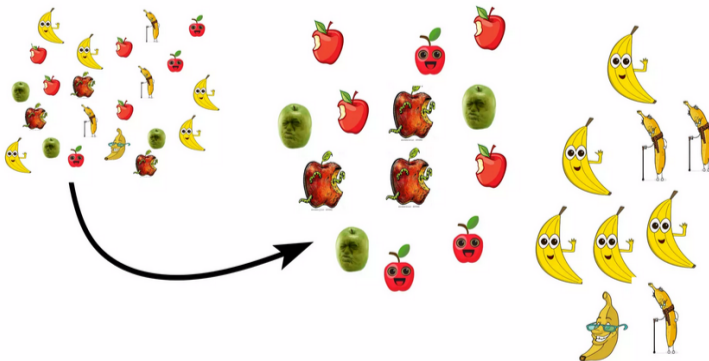


Supervised Learning: Classification



Unsupervised Learning: Clustering

How to **put** the fruits in **two** buckets?



Reinforcement Learning

- Close to human learning.
- Algorithm learns a policy of how to act in a given environment.
- Every action has some impact in the environment, and the environment provides rewards that guides the learning algorithm.

Reinforcement Learning

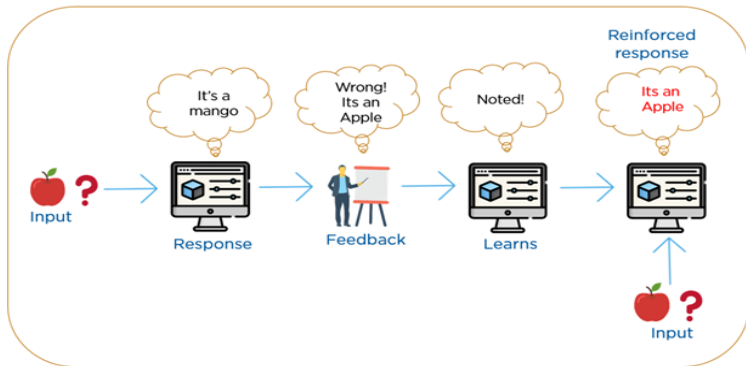


Figure: Working of reinforcement learning.

Elements of Reinforcement Learning

- Reward function
 - Defines the goal in an RL problem policy is altered to achieve this goal
- Value function
 - Reward function indicates what is good in an immediate sense while a value function specifies what is good in the long run.
 - Value of a state is the total amount of reward an agent can expect to accumulate over the future, starting from the state.
- Model of the environment
 - Predict mimic behavior of environment.
 - Used for planning and if know current state and action then predict the resultant next state and next reward.

IoT Use Cases Empowered by ML

Use cases

- **Energy Optimization:** Dynamically adjusting power usage in smart homes.
- **Anomaly Detection:** Identifying unusual patterns in sensor data.
- **Real-time Monitoring:** Monitoring of power consumption in smart grids.
- **Optimizing IoT network traffic:** Load balancing on servers.

Energy Optimization

■ Smart homes: respond to a user with a Smartphone

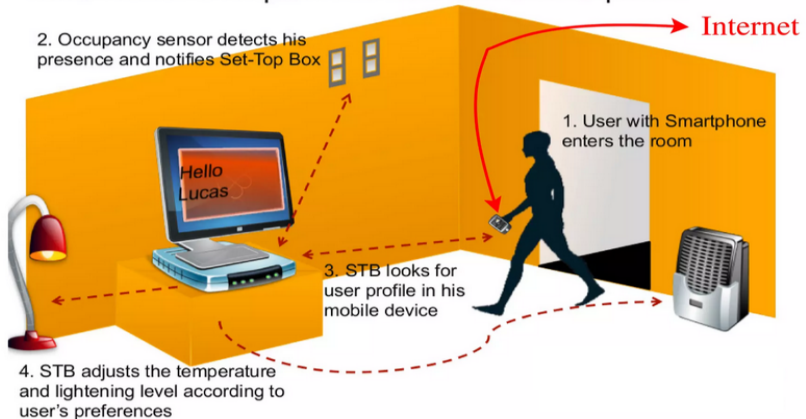


Image source: DigiHome platform

Anomaly Detection

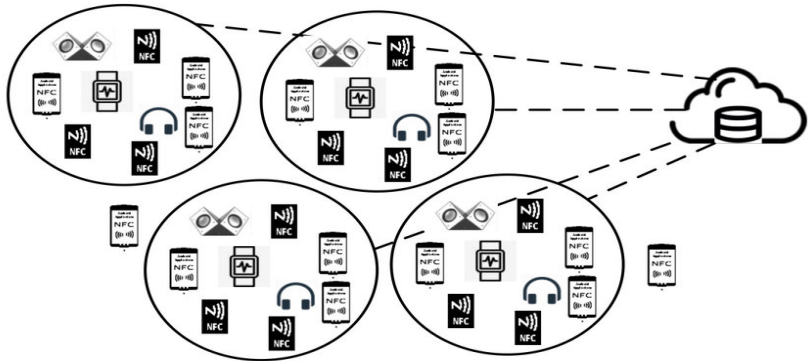


Figure: Identifying anomaly IoT device.

Real-time Monitoring

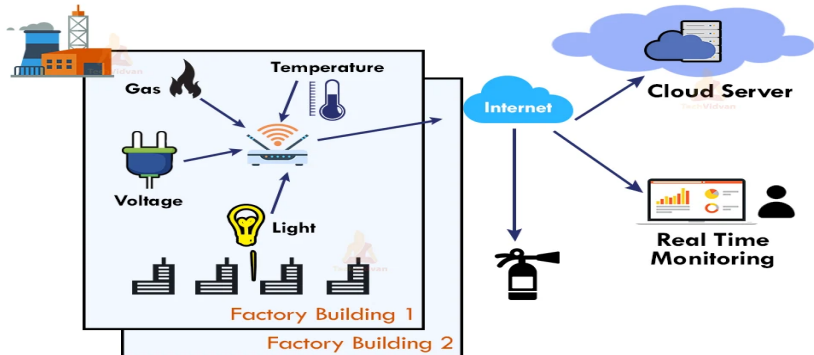


Figure: Monitoring of power consumption.

Optimizing IoT Network Traffic

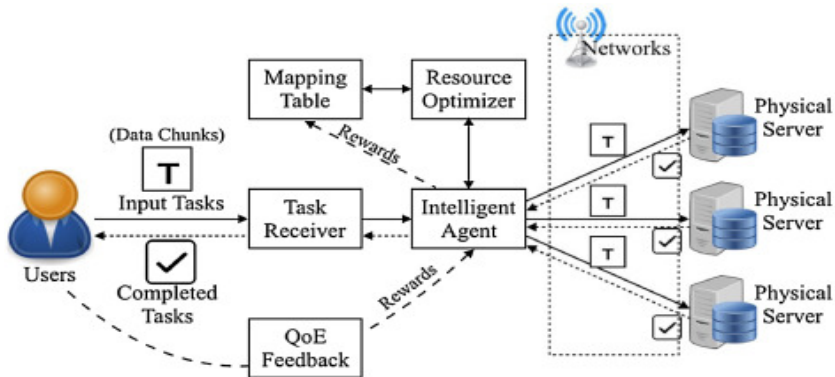


Figure: Load balancing on servers.

Challenges in Integrating ML with IoT

Challenges

- **Computational Limitations:** IoT devices have limited processing capabilities.
- **Real-Time Constraints:** ML models must process streaming data instantaneously.
- **Data Privacy:** Addressing data security regulations like GDPR.

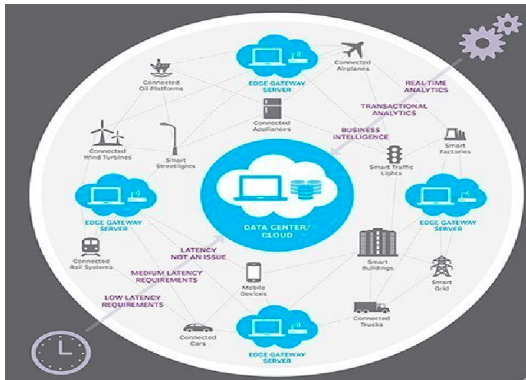
ML Architectures for IoT

Architecture

- **Edge Computing:** Processing data near its source for reduced latency.
- **Cloud Computing:** Centralized processing for computationally intensive tasks.
- **Federated Learning:** Decentralized training of models using data on devices.

Edge Computing

Any computing and network resources along the path between data sources centres



Edge Computing

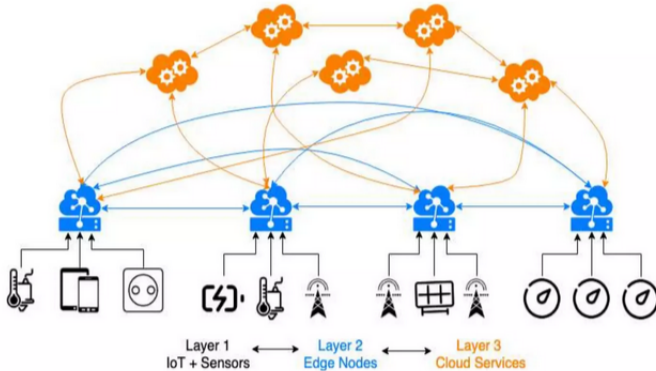


Figure: Edge computing architecture.

Edge Computing Characteristics

Characteristics

- Geographically distributed
- Bandwidth efficiency
- Autonomous
- Contextual and low latency
- Real time interaction
- Heterogeneous

Cloud Computing

Definition

Cloud computing provides the infrastructure, platforms, and software necessary for IoT devices to process and store massive volumes of data, enabling real-time analytics, scalability, and seamless communication.

Key Benefits:

- **Scalability:** Dynamically handle growing data from IoT devices.
- **Data Storage:** Centralized storage for massive IoT data streams.
- **Centralized Management:** Easier to manage and update compared to distributed systems.
- **Interconnectivity:** Enable seamless integration between diverse IoT devices.

Cloud Computing

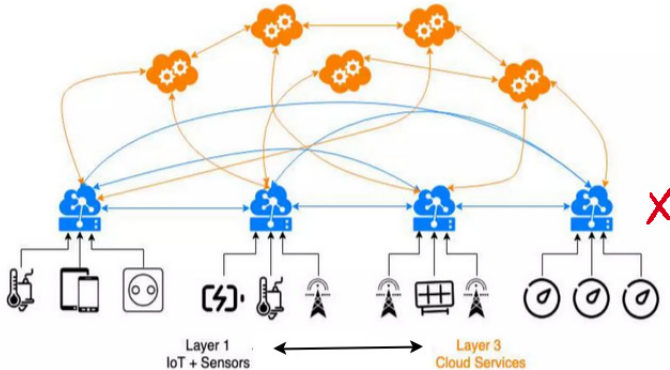


Figure: Cloud computing architecture.

Federated Learning

Definition

Federated learning” is a Machine Learning technique that moves models to the data instead of moving data to the models.

Key Points:

- Decentralized Data: Training data resides on distributed devices rather than being centralized in a single location.
- Model Training:
 - Each device trains a local model using its data.
 - Local updates are shared with a central server (not the raw data), which aggregates the updates to improve the global model.
- Privacy Preservation: Raw data never leaves the device.

Federated Learning

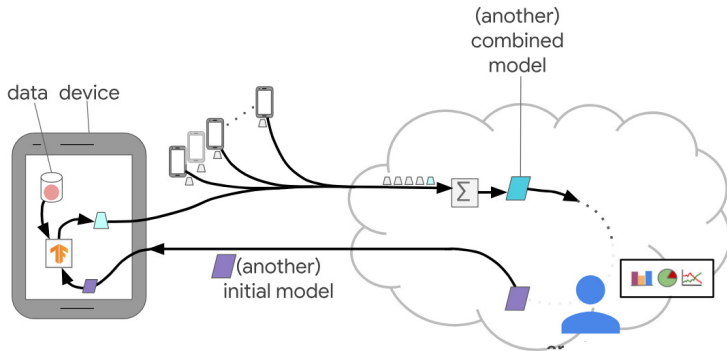


Figure: Federated Learning.

Federated Learning Workflow

- **Global Model Initialization:** The server initializes a global model and sends it to participating devices.
- **Local Training: Each device:**
 - Uses its local data to update the model.
 - Optimizes the model using a local objective function (e.g., stochastic gradient descent).
- **Model Update:**
 - The locally trained model parameters (gradients or weights) are sent back to the server.
 - Communication is often sparse or periodic to reduce overhead.
- **Aggregation:** The server aggregates local updates to update the global model.
- **Repeat:** Steps are repeated until convergence or a specified stopping criterion is met.

Applications of ML in IoT

Application

- **Smart Cities:** Traffic management, energy optimization.
- **Industrial IoT:** Robotics, predictive maintenance.
- **Healthcare:** Remote patient monitoring, wearable analytics.
- **Agriculture:** Precision farming, automated irrigation.

Research Trends and Challenges in IoT with ML

Research Trends

- **Edge AI:** On-device ML for real-time decision-making with low latency.
- **Explainable AI (XAI):** Ensures transparency in ML decisions, crucial for critical applications.
- **Autonomous IoT Systems:** Reinforcement learning for self-adaptive networks.
- **IoT Security with ML:** Advanced models for intrusion detection and device authentication.

Research Trends and Challenges in IoT with ML

Challenges

- **Data Privacy and Security:** Protecting sensitive IoT data during training and communication.
- **Resource Constraints:** Deploying lightweight ML models on power-limited devices.
- **Scalability and Interoperability:** Managing large-scale, heterogeneous IoT networks.
- **Data Quality:** Addressing noisy or imbalanced data from IoT devices.
- **Real-Time Processing:** Achieving low-latency decisions for high-speed data streams.

Future Directions in IoT with Machine Learning

- **Advanced Edge AI:**

- Ultra-efficient ML algorithms for resource-constrained IoT devices.
- Real-time processing with frameworks like TinyML.

- **AI and Blockchain Integration:**

- Secure, decentralized IoT data sharing with blockchain.
- Enhanced trust and fraud detection using ML.

- **Context-Aware Systems:**

- Personalized IoT systems incorporating contextual data.
- Applications in healthcare, smart homes, and industrial IoT.

- **Zero-Touch IoT Networks:**

- Autonomous network configuration, optimization, and repair.
- Powered by reinforcement learning.

Conclusion

Conclusions

- Machine learning enhances IoT systems by enabling data-driven insights and automation.
- The future of Machine Learning with IoT is virtually unlimited due to advances in technology and consumers desire to integrate devices.
- Challenges exist, but advances in ML techniques like federated learning and edge computing offer promising solutions.

Questions?

Thank you!
Questions and Discussion